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Crater morphology of primordial black hole impacts

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ABSTRACT

In this work, we propose a novel campaign for constraining relativistically compact massive compact halo object (MACHO) dark matter, such as primordial black holes (PBHs), using the Moon as a detector. PBHs of about 10^{19} to 10^{22} g may be sufficiently abundant to have collided with the Moon in the history of the Solar system. We show that the crater profiles of a PBH collision differ from traditional impactors and may be detectable in high-resolution lunar surface scans now available. Any candidates may serve as sites for *in situ* measurements to identify high-pressure phases of matter which may have formed near the PBH during the encounter. While we primarily consider PBH dark matter, the discussion generalizes to the entire family of MACHO candidates with relativistic compactness. Moreover, we focus on the Moon since it has been studied well, but the same principles can be applied to other rocky bodies in our Solar system without an atmosphere.

Key words: black hole physics – hydrodynamics – shock waves – methods: numerical – Moon – dark matter.

1 INTRODUCTION

One long-standing question in astrophysics and cosmology is the nature of dark matter (DM; e.g. Bertone & Hooper 2018, and reference therein). Candidates for DM particles can be divided into two categories: microscopic and macroscopic, where the latter are often referred to as massive compact halo objects (MACHOs). It has been suggested that DM is composed (at least in part) of black holes (BHs) formed from density fluctuations in the early Universe (Carr, Kühnel & Sandstad 2016). Such objects are called primordial black holes (PBHs). Since the idea has been proposed, different observations have excluded certain mass ranges for these PBHs (Carr et al. 2020; Carr & Kühnel 2020). Currently, there are several poorly constrained mass windows, notably between 10^{17} and 10^{23} g, where the constraints are weak enough that PBHs could be a large fraction of the DM , though the exact mass windows are dependent on the PBH mass spectrum. While several constraints have been reported in this mass range some have been contested (Katz et al. 2018), which motivates this work to develop a new independent constraint in this asteroid-mass to sublunar-mass window.

In this study, we consider the possibility that a PBH collided with the Moon in the past. As we shall show later, the craters from such a collision will have a shape markedly different from an ordinary meteor impact. This feature can readily be detected by surface images and radar scans.

The idea of a collision between a PBH (or another compact object) with planetary bodies has been considered by various authors in the past. Rafelski et al. (2013) have considered in some detail the physics of compact ultra-dense matter colliding with Solar system bodies. They found that relativistically compact matter will generally completely penetrate the body forming both 'entrance' and 'exit' wounds on a hyperbolic trajectory. Penetration is generally supersonic which may produce a seismic signal (Khriplovich et al. 2008) which has been searched for (Herrin, Rosenbaum & Teplitz 2006; Banerdt et al. 2007; Horowitz & Widmer-Schnidrig 2020). Accretion on to a high-velocity PBH 'entering' the Earth has been proposed as an explanation for the Tunguska event (Jackson & Ryan 1973). 'Exit' events have meanwhile been compared to volcanic events, transporting crust material high into the atmosphere (Rafelski et al. 2012, 2013). While surface processes on the Earth complicate efforts to establish historic events as PBH encounters, the Moon and other Solar system bodies without active surface processes hold a record of several billion years of impacts in their surface geology. In this work, we develop the theory of cratering dynamics in PBH encounters with solar bodies and describe a possible observational campaign to constrain low-mass MACHO DM .

While we generally consider PBHs interacting with the Moon in this work, our discussion generalizes to many Solar system bodies and ultra-dense DM candidates.

The structure of the paper is as follows. In Section 2, we present numeric hydrodynamic simulations that compare the evolution of the ejecta for normal and PBH collisions. In Section 3, we present analytic calculations of the shock wave trajectories in both cases. Using the shock trajectories, we calculate how the ejecta will be distributed outside the crater, and show that each one of these cases will yield a unique ejecta distribution. Finally, in Section 4, we discuss the results and the implications for PBHs as DM particles.

2 NUMERICAL SIMULATIONS

We simulate impact events using the moving mesh, hydrodynamic code RICH (Yalinewich et al. 2015). In Fig. 1, we present sequential log density snapshots from two simulations, described below, to highlight the differences between the regular and PBH impacts.

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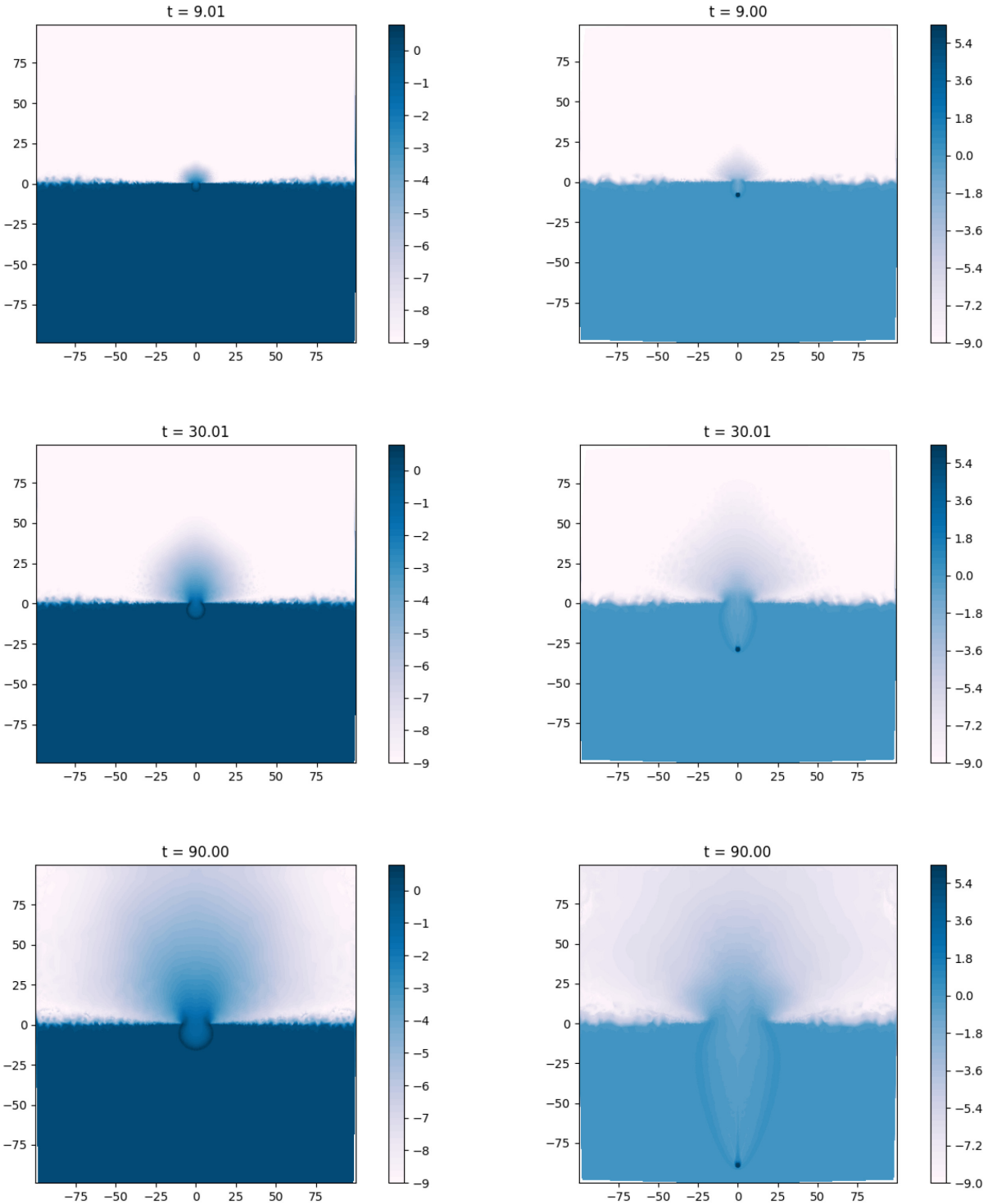

Figure 1. Log density snapshots from numerical simulations of a regular, head-on impact (left), and an impact of a PBH (right), both without gravity. Both impactors are of the same size and initially move at the same velocity. The density of the regular impactor is the same as that of the ground, while the density of the PBH is six orders of magnitude higher. The distances shown in the images are normalized by the radius of the impactor. The time is normalized by the time it takes the impactor to traverse a distance equal to its radius. The density is normalized to the density of the target. These snapshots show that while the regular impactor is decelerated quickly and deposits its energy in the target, the PBH keeps ploughing through the target. It is also apparent that the distribution of ejecta is different.

In the first simulation, we collide a cold sphere with a cold half-space, both filled with ideal gas with the same density. This simulation is supposed to represent a normal cratering event. We chose the adiabatic index to be $\gamma = 1.4$, since this value best describes the high-pressure shock behaviour in silica (Yalinewich & Schlichting 2019). Since we want to resolve the impactor and also

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follow the shock wave to distances much larger than the impactor radius, we arranged 20 000 mesh generating points in a non-uniform way inside the computational domain. In the vicinity of the impactor, the resolution is 0.1 of the radius of the impactor, but at larger radii the cells are larger such that the ratio between the cell size and radius is roughly constant around 0.1. With this distribution of points, we were able to extend the boundaries of the computational domain to 10^6 impactor radii. To simplify the simulation, we assume a head-on collision, and run the simulation on a 2D grid using cylindrical symmetry. The density of both the impactor and the target is set to one while the density in the rest of the computational domain is 10^{-9}, the velocity with which the impactor travels towards the target is set to one and the ambient pressure before impact is 10^{-9} everywhere. We ran the simulation to time 10^7, where the unit of time here is the time it takes the impactor to traverse its own radius. The simulation does not include gravity, so the debris expands ballistically from the impact site. In Fig. 2, we plotted the density of the ejecta as a function of distance from the impact site, and find the density slope to be $\mathrm{dln}\,\rho/\mathrm{dln}\,r = -4.1$.

The second simulation is intended to describe a head-on collision of a PBH with the surface of the Moon. The setup for this simulation is the same that described in the previous paragraph, except the density of the impactor is 10^6 times greater than that of the target, and also we increase the resolution along the trajectory of the impactor inside the target, so the total number of points is larger (120 000). We note that a PBH effective density is even larger than the value used here, but as will be discussed in the next section, above a certain threshold the exact value of the impactor density does not matter because for sufficiently high impactor densities (relative to the target) the impactor does not decelerate appreciably during the encounter. This approximation is valid so long as the column of matter swept up by the impactor is much less massive than the impactor, which is the case in our simulation. The radial density profile from this simulation is shown in Fig. 3. From this figure, we infer the slope $\mathrm{dln}\,\rho/\mathrm{dln}\,r = -5.9$.

3 THEORETICAL MODELLING

3.1 Classical cratering

We begin by describing classical craters, which form as a result of meteor impacts or surface explosions. In the former, an impactor with mass m_i collides with ground at some velocity v_i. For simplicity, we assume that both the impactor and the ground have the same density ρ_0. As a result of the collision, a shock wave emerges from the impact site and travels into the soil. As the shock wave expands, it goes through few distinct phases. In the first phase, while the distance travelled by the shock is much smaller than the radius of the impactor, the shock velocity is roughly uniform and roughly equal to v_i. In the second stage, the shock begins to decelerate as it sweeps up more mass. As the shock expands, it loses information about the impactor's incidence angle, and converges to the same universal solution (Yalinewich & Remorov 2020). For this reason, we can just consider head-on collisions. In the case of hypervelocity impacts, the velocity decelerates according to some power law in the swept up mass $v \approx v_i(m/m_i)^{-\beta}$ where β is some constant. Energy and momentum conservation provide limits on the value of this power-law index (Zeldovich 2015) $1 > \beta > 1/2$. The value of β depends on the equation of state of the soil (Housen et al. 1983; Yalinewich & Remorov 2020), but is typically close to 2/3. As the shock excavates a crater basin, material is expelled out of the mouth of the basin. A fluid element emerges from the basin with roughly the same velocity with which it was shocked. The expelled material eventually settles outside the rim of the crater, forming the so-called ejecta blanket (Osinski et al. 2011). The process by which an ejecta blanket forms is referred to as ballistic sedimentation. The final radius of debris expelled at a velocity v is $r \approx v^2/g$, where g is the gravitational acceleration. We can use the velocity profile of the shock to calculate the surface density profile of the ejecta blanket

$$\sigma \approx \frac{m}{r^2} \approx r^{-2-1/2\beta}. \tag{1}$$

Energy and momentum conservation yield limits on the slope of the surface density distribution $-2.5 > \mathrm{dln}\,\sigma/\mathrm{dln}\,r > -3$. If we assume a constant density for the settled ejecta, then the thickness of the ejecta blanket scales the same way as the surface density. Most ejecta blankets around craters on the Moon are within this range (Zhu et al. 2015, 2017; Raggio et al. 2016). In contrast, ejecta blankets from chemical and nuclear explosions on Earth are slightly steeper $-3 > \mathrm{dln}\,\sigma/\mathrm{dln}\,r > -4$, due to air drag (Carlson & Jones 1965).

Using the same shock trajectory we can make a theoretical prediction for the density slope we numerically calculated in the previous section. We saw that $v \propto m^{-\beta}$. At distances much larger than the radius of the impactor, the velocity profile is roughly homologous $v \approx d/t$, where d is the distance from the impact site and t is the time since the impact. Combining these relations with the definition of density $m \approx \rho d^3$ we find a scaling relation for the density profile $\rho \propto r^{-3-1/\beta}$. For $\beta = 2/3$ we obtain $\mathrm{dln}\,\rho/\mathrm{dln}\,r = -4.5$, which is close to the value obtained in the numeric simulation $\mathrm{dln}\,\rho/\mathrm{dln}\,r = -4.1$.

Eventually, the shock stops and the crater reaches its final size. There are two criteria that can determine the final size of the crater (which will always be the smaller of the two). One possibility is that the shock stops when the shock pressure drops below the elastic modulus of the soil Y, so the final shock radius is

$$R \approx R_i \left(\rho_0 v_i^2/Y\right)^{1/6\beta}, \tag{2}$$

where $R_i \approx (m_i/\rho_0)^{1/3}$ is the radius of the impactor. The second possibility is that the shock stops when the shock pressure drops below the hydrostatic pressure $g\rho R$, and so

$$R \approx R_i \left(v_i^2/gR_i\right)^{1/(6\beta+1)}. \tag{3}$$

In the case of the lunar regolith, this is the deciding condition.

3.2 Craters from PBHs

The density of PBHs is much higher than that of meteors, so instead of transferring all of their energy to the target at the impact site, the impactor just passes through and deposits roughly the same amount of energy per unit length along its trajectory inside the target. For this reason, above some critical threshold, the evolution of the shock wave is independent of the density of the impactor. To understand how the shock wave will behave in this case, one can think about this problem as a series of point explosions, where the energy of each explosion scales with its depth $E \propto x$. If the mass of the BH is m_i, then the effective cross-section is determined by the Bondi radius $R_b \approx Gm_i/v_i^2$, and so $E \approx xR_b^2 v_i^2 \rho$. For a uniform density soil, the mass in a sphere of radius x around the hotspot is $m \approx \rho_0 x^3$. Hence, $v \approx v_i R_b \rho_0^{1/3} m^{-1/3}$. We thus obtain a new power-law relation between the shock velocity and the swept up mass, which is shallower than that obtained for classical craters, i.e. $\beta = 1/3$. Using the same reasoning as in the previous section, we find the slopes of

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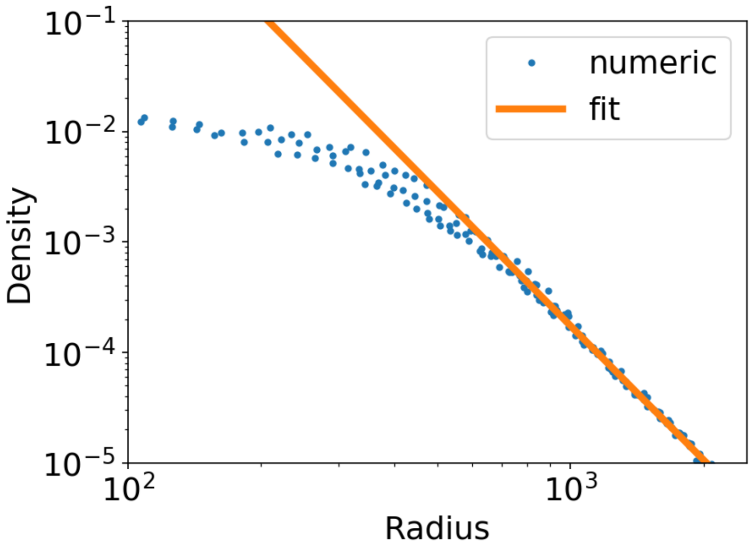

Figure 2. Radial density profile for a normal cratering event, obtained from numerical simulations. The radius is given in units of the impactor radius, and the density in units of the target (and impactor) density. The slope of the power-law fit is -4.1, whereas the analytic prediction is -4.5.

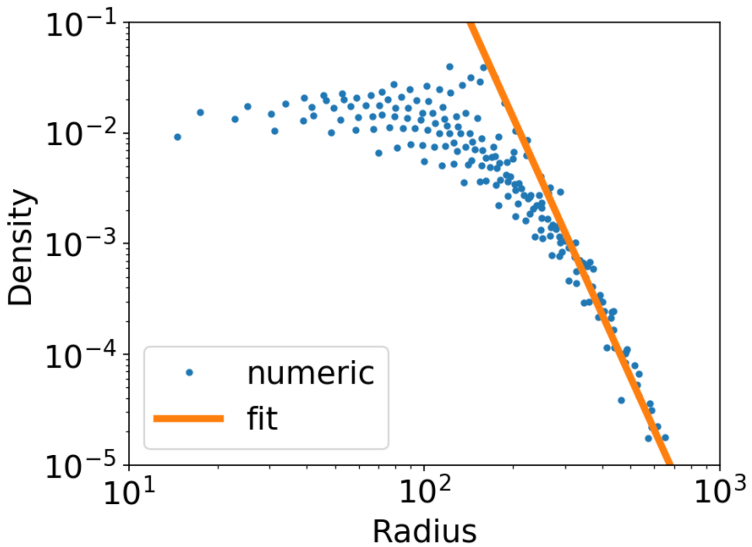

Figure 3. Radial density profile for a head-on PBH collision, obtained from numerical simulations. The radius is given in units of the impactor radius and the density in units of the target density. The slope of the power-law fit is -5.9, whereas the analytic prediction is -6. The numerical simulation confirms the analytic prediction.

ejecta blanket surface density profile

$$\frac{\mathrm{d}\ln\sigma}{\mathrm{d}\ln r} = -3.5. \tag{4}$$

As was done in the previous section, we can also calculate the density distribution of the ejecta. Our analytic theory predicts that $\beta = 1/3$, and so the slope of the density profile should be $\mathrm{d}\ln\rho/\mathrm{d}\ln r = -6$, which is very close to the value obtained from the numerical simulation $\mathrm{d}\ln\rho d/\mathrm{d}\ln r = -5.9$.

To find the size of the crater, we need to find the depth at which when the shock reaches the surface, the shock velocity is comparable to the free fall velocity from the top of the basin to the bottom. The final crater radius is therefore

$$R \approx \left(\frac{v_i^2 R_b^2}{g}\right)^{1/3} \approx 1.9\left(\frac{v_i}{200\,\frac{\mathrm{km}}{\mathrm{s}}}\right)^{2/3} \times \left(\frac{m_i}{10^{19}\,\mathrm{g}}\right)^{2/3}\left(\frac{g}{2\,\frac{\mathrm{m}}{\mathrm{s}^2}}\right)^{-1/3}\mathrm{m}. \tag{5}$$

Craters of this size could be detected, for example, with the Lunar Reconnaissance Orbiter, which has a resolution of about 1-m (Robinson et al. 2010). Hence, any interaction with BHs more massive than 10^{19} g could, in principle, be detected. The number of such collisions in the lifetime of the Solar system is

$$N \approx 0.1\left(\frac{\rho_{DM}}{8\cdot 10^{-25}\,\mathrm{g\,cm^{-3}}}\right)\left(\frac{m_i}{10^{19}\,\mathrm{g}}\right)^{-1}\left(\frac{v_i}{200\,\mathrm{km\,s^{-1}}}\right), \tag{6}$$

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where ρ_{DM} is the density of DM . While this does not necessarily guarantee a detectable encounter between the moon and a PBH in the history of the Solar system, it is of sufficiently high probability to be interesting. Furthermore, if the Solar system has passed through regions of higher DM density in the past the expected number of events could be higher.

4 CONCLUSIONS

In this work, we considered the possibility that a PBH collided with the Moon. We show that a crater resulting from such a collision will have a different shape from a regular meteor impact, namely, a compact object impact crater will have steeper ejecta blanket. Most craters on the Moon have ejecta blankets whose shape is compatible with a meteor impact. However, there are craters with steeper ejecta blankets whose origin does not involve exotic objects (e.g. Mare Imbrium, see Zhu et al. 2017).

Steep ejecta blankets can therefore be used to flag craters of interest, but are not indisputable evidence of a compact object impact. A stronger evidence for BH impact is the presence of a high pressure, pyrite-type phase of silica (Kuwayama et al. 2005). Regular impacts on the Moon only produce the stishovite and seifertite phases of silica. This is because the new phase of silica appears above 40 TPa, whereas peaks pressure in a typical collision is around 1 TPa (Kayama, Nagaoka & Niihara 2018). We therefore propose that steep ejecta blankets can be used to flag craters of interest, but to confirm a BH impact origin would require finding the high-pressure phase of silica.

If such a crater did exist on the Moon, there is a possibility that it will have eroded away, due to meteor impacts or solar wind sputtering. For a 1-m crater, the erosion time is over a billion years (McDonnell et al. 1977), and for larger craters even longer. Hence, we conclude that erosion is not a concern. We note that the trail of shocked material underneath the crater can provide further evidence for a PBH origin, but finding it would require excavation, or non-intrusive methods such as seismic refraction or ground-penetrating radar.

The smallest BHs that produce a detectable 1-m crater have a mass of about 10^{19} g. We estimated that even for these small BHs, the impact rate is so low that, even if all DM is composed of such BHs, there's only about a 10 per cent chance that one collided with the Moon throughout its lifetime. With such a low rate we are not able to place constraints on the DM distribution. However, the same considerations used here for the Moon could apply just as well to any rocky body without an atmosphere. Therefore, with future missions to other rocky bodies in our Solar system it will eventually be possible to use the method outlined in this work to constrain the mass distribution of PBHs .

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DATA AVAILABILITY

The source code for the numerical simulation and documentation can be found on github at https://github.com/bolverk/huji-rich.

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